

OESOPHAGUS

Regular anatomy notes - course, constrictions, parts, relations, supply, structure and applied anatomy

Core idea

The **oesophagus** is a narrow muscular tube connecting the pharynx to the stomach. It is mainly a conduit for food and liquids during swallowing.

It is about **25 cm long**, normally collapsed anteroposteriorly, and opens only during passage of a food bolus.

For exams, learn it under five heads: **extent and course, constrictions, relations, neurovascular supply, and applied anatomy.**

Must-remember facts

Begins: lower border of cricoid cartilage, opposite C6.

Ends: cardiac orifice of stomach, around T11.

Pierces diaphragm: T10 through the oesophageal hiatus.

Lower end: important site of portocaval anastomosis.

Parts and lengths

Cervical part: about 4 cm.

Thoracic part: about 20 cm.

Abdominal part: about 1-2 cm; shortest part.

Total length: about 25 cm.

1. Dimensions, lumen and extent

Feature	Important points
Length	About 25 cm, roughly 10 inches.
Width	About 2 cm.
Lumen	Flattened anteroposteriorly; normally collapsed and opens during swallowing.
Upper end	Continuation of pharynx at lower border of cricoid cartilage, opposite lower border of C6 vertebra.
Lower end	Cardiac orifice of stomach at about T11, around 2.5 cm left of the median plane.
Diaphragmatic opening	Pierces diaphragm at T10 through the oesophageal hiatus.

Functional explanation

The oesophagus is built for **transport, not digestion**. Its collapsed lumen prevents unnecessary air entry, while its muscular wall produces peristaltic waves to push the bolus toward the stomach.

Because it lies behind the trachea in the neck and behind the heart in the thorax, enlargement of nearby structures can compress it and produce **dysphagia**.

2. Course and curvatures

Course: The oesophagus begins at the lower border of the cricoid cartilage, descends in front of the vertebral column, passes through the superior and posterior mediastina, pierces the diaphragm at T10, enters the abdomen and opens into the stomach at the cardiac orifice around T11.

Side-to-side curvatures

First left deviation: at the root of neck before entering the thoracic inlet.

Second left deviation: around T7, before passing in front of the descending thoracic aorta.

These deviations are important during oesophagoscopy and tube passage.

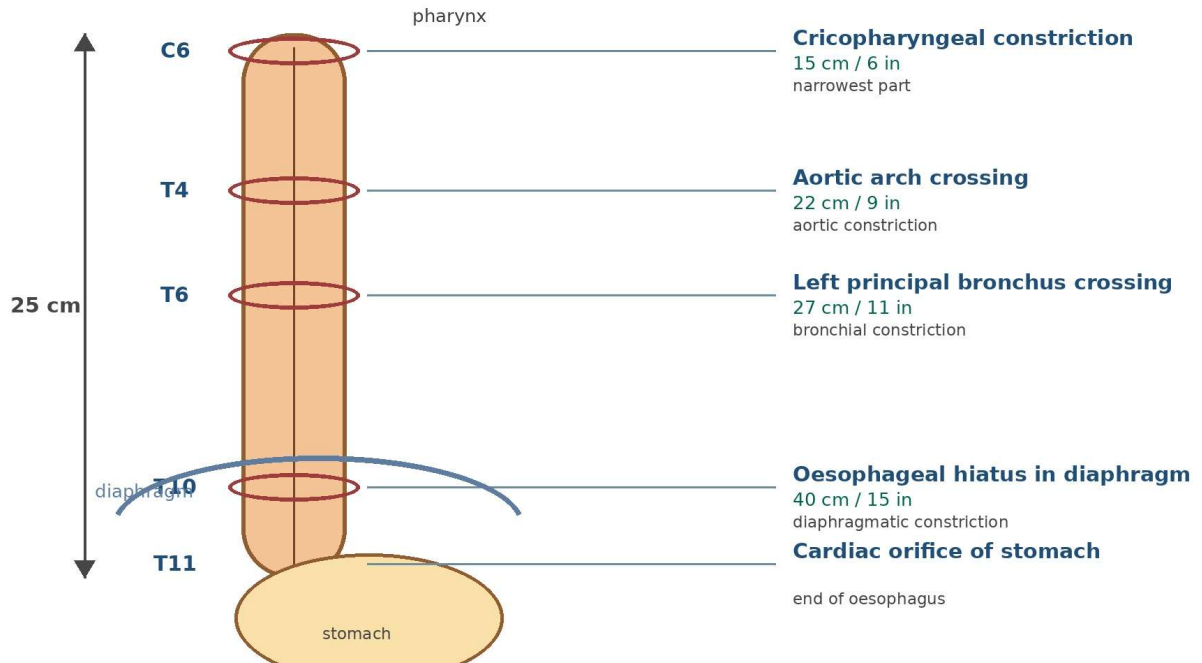
Anteroposterior curvatures

One follows the cervical spine.

Another follows the thoracic spine.

The tube is therefore not a perfectly straight vertical pipe.

Oesophagus: course and normal constrictions



3. Anatomical constrictions

Why constrictions matter

Normal constrictions are clinically important because swallowed foreign bodies commonly lodge there, corrosive strictures may form there, carcinoma has predilection for these sites, and tubes/endoscopes may be difficult to pass through them.

Constriction	Vertebral level	Distance from upper incisors	Clinical note
Pharyngo-oesophageal / cricopharyngeal	C6	15 cm / 6 inches	Narrowest part; common site for foreign body impaction.
Aortic constriction	T4	22 cm / 9 inches	Where crossed by arch of aorta.
Bronchial constriction	T6	27 cm / 11 inches	Where crossed by left principal bronchus.
Diaphragmatic constriction	T10	40 cm / 15 inches	At oesophageal hiatus of diaphragm.

Exam trap: Do not confuse vertebral levels with distances from incisor teeth. Levels are C6, T4, T6 and T10. Distances are 15, 22, 27 and 40 cm.

4. Parts of the oesophagus

Part	Extent	Length	Important notes
Cervical	Lower border of cricoid cartilage to superior border of manubrium sterni.	About 4 cm	Lies behind trachea; inclines slightly to the left; important in neck surgery.
Thoracic	Superior border of manubrium sterni to oesophageal opening in diaphragm.	About 20 cm	Passes through superior and posterior mediastina; has close relations with trachea, aorta, left bronchus, left atrium and vagus nerves.
Abdominal	Oesophageal opening of diaphragm to cardiac end of stomach.	About 1-2 cm	Shortest part; partly covered by peritoneum; important site of portocaval anastomosis.

Practical classification

The **cervical part** is important in neck surgery because it lies behind the trachea and is related to recurrent laryngeal nerves.

The **thoracic part** is important in mediastinal anatomy and radiology.

The **abdominal part** is clinically important because it is a major site of portocaval anastomosis and reflux-related disease.

5. Relations

A. Cervical part relations

Relation	Structures
Anterior	Trachea and recurrent laryngeal nerve.
Posterior	Prevertebral fascia, longus colli muscles and vertebral column.
On each side	Lobe of thyroid gland, common carotid artery and thoracic duct on the left side.

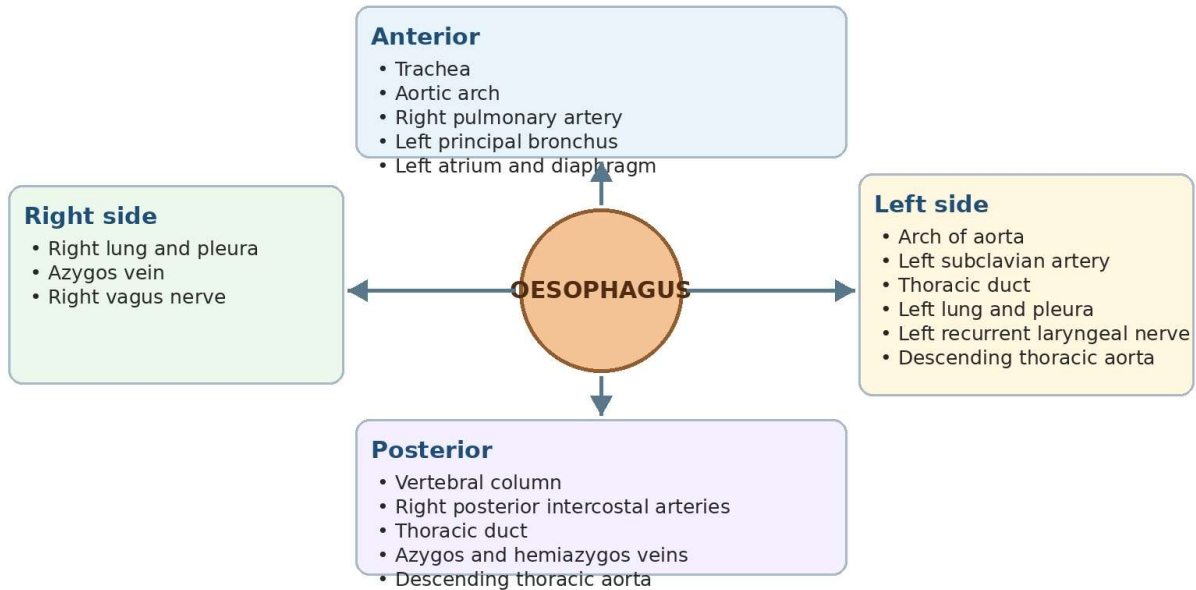
Surgical point

The left margin of the cervical oesophagus projects laterally from behind the trachea in the neck. Therefore, the cervical part can be exposed more easily from the left side.

The prevertebral fascia forms a movable base on which the trachea and oesophagus move up and down during swallowing and phonation.

B. Thoracic part relations

Thoracic oesophagus: relation map



Side	Relations
Anterior, from above downward	Trachea; arch of aorta; right pulmonary artery; left principal bronchus; left atrium enclosed in pericardium; diaphragm.
Posterior	Vertebral column; right posterior intercostal arteries; thoracic duct; azygos vein; terminal parts of hemiazygos veins; descending thoracic aorta.
Right side	Right lung and pleura; azygos vein; right vagus nerve.
Left side	Arch of aorta; left subclavian artery; thoracic duct; left lung and pleura; left recurrent laryngeal nerve; descending thoracic aorta.

How to remember thoracic relations

Think of the oesophagus as the **posterior food tube**. In front are airway, aortic arch, bronchus and heart. Behind are vertebral column, thoracic duct, azygos system and thoracic aorta. On the sides it touches pleura/lungs, vagus nerves and major vessels.

C. Abdominal part relations

Relation	Structures / notes
Course	Enters abdomen through oesophageal opening of diaphragm, runs downward and to the left in front of left crus of diaphragm, behind left lobe of liver, and opens into cardiac end of stomach.
Anterior	Anterior vagal trunk and left lobe of liver.
Posterior	Posterior vagal trunk and diaphragm / left crus.
Peritoneum	Covered by peritoneum anteriorly and on the left side; it is the only part with serous covering.
Vessels along left side	Oesophageal branches of left gastric artery and accompanying veins.

6. Arterial supply, venous drainage and lymphatic drainage

Part	Arterial supply	Venous drainage	Lymphatic drainage
Cervical	Inferior thyroid arteries.	Inferior thyroid veins and left brachiocephalic vein.	Deep cervical and pretracheal lymph nodes.
Thoracic	Oesophageal branches of descending thoracic aorta and bronchial arteries.	Azygos and hemiazygos veins.	Posterior mediastinal lymph nodes.
Abdominal	Oesophageal branches of left gastric and left inferior phrenic arteries.	Hemiazygos vein and left gastric vein.	Left gastric nodes; lower third may drain to coeliac nodes.

Pattern to remember

The supply follows the part of the tube: **neck from inferior thyroid vessels, thorax from thoracic aorta/bronchial vessels, and abdomen from left gastric/inferior phrenic vessels.**

The lower end has both systemic and portal venous drainage, making it an important site of **portocaval anastomosis**.

7. Nerve supply and referred pain

Parasympathetic supply

Derived from recurrent laryngeal nerves and oesophageal plexuses formed by vagus nerves.

Functions: sensory, motor and secretomotor supply.

At hiatus: left vagus forms anterior gastric nerve/trunk; right vagus forms posterior gastric nerve/trunk.

Sympathetic supply

Derived mainly from T5-T9 spinal segments.

Functions: sensory and vasomotor.

Pain may be referred to lower thoracic and epigastric regions.

Referred pain: Pain sensations commonly arise from the lower oesophagus, which is vulnerable to acid-peptic oesophagitis. The pain is carried mainly by sympathetic fibres and may be referred to the lower thoracic and epigastric regions, sometimes mimicking cardiac pain.

8. Structure of the wall

Wall of the oesophagus - layers and muscle change

Mucosa

Stratified squamous non-keratinized epithelium; lamina propria; thick muscularis mucosae

Submucosa

Contains mucous oesophageal glands

Muscular layer

Upper 1/3 skeletal; middle 1/3 mixed; lower 1/3 smooth

Adventitia

Dense connective tissue with elastic fibres

Upper 1/3
skeletal

Middle 1/3
mixed

Lower 1/3
smooth

towards stomach

Layer	Key points
Mucosa	Epithelium is stratified squamous non-keratinized. Lamina propria is present. Muscularis mucosae is very thick and contains longitudinal smooth muscle fibres.
Submucosa	Contains mucous oesophageal glands.
Muscular layer	Upper one-third: skeletal muscle. Middle one-third: both skeletal and smooth muscle. Lower one-third: smooth muscle.
Adventitia	Dense connective tissue with many elastic fibres. Most of the oesophagus lacks serosa; the abdominal part has peritoneal covering.

Functional meaning of wall structure

Protective stratified squamous epithelium resists abrasion by food. Mucous glands lubricate the lumen. The change from skeletal to smooth muscle explains the transition from voluntary initiation of swallowing to involuntary peristalsis.

9. Development

Origin: The oesophagus develops from the foregut. The respiratory tract develops from a foregut diverticulum called the laryngotracheal diverticulum/tube.

Developmental event	Applied importance
Separation of laryngotracheal tube by formation of laryngotracheal septum.	Defective separation can produce tracheo-oesophageal fistula.
Recanalization of temporarily obliterated oesophageal lumen.	Failure of recanalization can produce oesophageal atresia.

Common congenital association

In the common type of tracheo-oesophageal fistula, the upper oesophagus ends blindly while the lower oesophagus communicates with the trachea. Clinical clues include hydramnios, air-filled stomach and coughing/vomiting during feeding.

10. Applied anatomy

Condition / procedure	Anatomical basis
Dysphagia	Difficulty in swallowing can occur due to external compression by aortic aneurysm, enlarged lymph nodes, abnormal right subclavian artery, enlarged left atrium, or by luminal narrowing from stricture/carcinoma.
Barium swallow	Shows normal impressions produced by arch of aorta, left principal bronchus and left atrium. It is used to assess strictures, carcinoma, achalasia and left atrial enlargement.
Oesophagoscopy / gastric tube passage	Normal constrictions must be remembered to avoid difficulty or injury while passing instruments.
Achalasia cardia	Failure of lower oesophageal sphincter to relax during swallowing causes food accumulation and regurgitation; barium swallow may show bird-beak / rat-tail appearance.
Oesophageal varices	Portal hypertension opens and dilates portocaval channels at the lower oesophagus; rupture causes haematemesis.
Barrett oesophagus	Stratified squamous epithelium is replaced by gastric-type epithelium; it may predispose to oesophageal carcinoma.
Malignant tumours	Commonly occur in the lower one-third; lymph may pass below diaphragm to coeliac nodes, so spread can be extensive.
Mitral stenosis	Enlarged left atrium may compress the oesophagus causing dysphagia; it may also affect nearby recurrent laryngeal nerve producing hoarseness.

11. Long-answer framework for exams

How to write an answer on oesophagus

1. Start with definition, length, extent and parts.
2. Draw a simple vertical diagram and mark C6, T4, T6, T10 and T11.
3. Add constrictions with distances from upper incisors.
4. Write relations part-wise: cervical, thoracic and abdominal.
5. Add blood supply, venous drainage, lymphatics and nerve supply.
6. Finish with applied anatomy: dysphagia, varices, achalasia, barium swallow, tracheo-oesophageal fistula and carcinoma.

Diagram labelling checklist

- ☐ Pharyngo-oesophageal junction / cricopharyngeus
- ☐ Arch of aorta
- ☐ Left principal bronchus
- ☐ Oesophageal hiatus of diaphragm
- ☐ Cardiac orifice of stomach
- ☐ Thoracic duct
- ☐ Azygos and hemiazygos veins
- ☐ Vagus nerves / oesophageal plexus
- ☐ Left atrium and pericardium
- ☐ Descending thoracic aorta

12. Revision sheet

Quick viva question	Answer to remember
Length of oesophagus?	About 25 cm.
Beginning and ending levels?	Begins at C6; ends at cardiac orifice of stomach around T11.
Level of oesophageal hiatus?	T10.
Parts?	Cervical, thoracic and abdominal.
Shortest part?	Abdominal part, about 1-2 cm.
Narrowest part?	Commencement at cricopharyngeal sphincter.
Constriction distances?	15, 22, 27 and 40 cm from upper incisors; or 6, 9, 11 and 15 inches.
Normal barium swallow impressions?	Arch of aorta, left principal bronchus and left atrium.
Site of portocaval anastomosis?	Lower end / abdominal part of oesophagus.
Cause of oesophageal varices?	Portal hypertension causing dilated portosystemic channels.
Nerve supply?	Parasympathetic from vagus/recurrent laryngeal nerves; sympathetic from T5-T9.
Why left-sided approach in neck?	Cervical oesophagus projects slightly to the left from behind the trachea.

Last-minute memory hooks

- **C6 to T11** = total course.
- **T10** = oesophageal hiatus.
- **6, 9, 11, 15 inches** = constrictions from upper incisors.
- **First constriction** is the narrowest.
- **Lower end** = portocaval anastomosis and varices.
- **Lower one-third** = clinically important for reflux, Barrett change and carcinoma spread.

Source base

Prepared from: Vishram Singh, Textbook of Anatomy - Upper Limb and Thorax, Vol. I; Textbook of Anatomy - Head, Neck and Brain, Vol. III; and Textbook of Anatomy - Abdomen and Lower Limb, Vol. II.